External Defibrillator Vest

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Abstract

Now a days cardiac arrhythmias of ventricular fibrillation and ventricular tachycardia are very common diseases which require a fast treatment. People who suffer cardiac arrest need emergency medical treatment. CPR (cardiopulmonary resuscitation) can help maintain the flow of oxygen to the brain, but getting the heart restarted and working normally often requires defibrillation with an electric shock.

This project presents a design and prototype implementation of wearable defibrillator that automatically diagnoses the potentially life threatening arrhythmia, ventricular fibrillation and ventricular tachycardia in a patient and is able to treat it through defibrillation. The system determines when a shock is warranted, and delivers it along with an audio warning stating that a shock is about to be delivered so that the patient, if conscious, settles himself down or if needed, stops the device from delivering the shock by pressing a manual button .

The proposed system implements ECG monitoring and processing through Pan+Tompkins algorithm. The defibrillator is made light in weight to be carried around easily. The proposed system is made better than the commercially available defibrillator vests from cost and portability point of view.

Keywords: CPR, Wearable defibrillator, arrhythmias, ventricular fibrillation, ventricular tachycardia, Pan+Tompkins

UNDERTAKING

I certify that research work titled "**External Defibrillator Vest**" is our own work. The work has not been presented elsewhere for assessment. Where material has been used from other sources it has been properly acknowledged / referred.

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YaAllah,

Send your peace and blessings

On the final Prophet(P.B.U.H),

And his family,

And companions,

And those who follow him. {ameen}

ACKNOWLEDGEMENTS

All the praises are for Almighty Allah Who granted His blessings to us and made us capable of performing this task. We would like to express our deepest appreciation to all those who provided us the possibility to complete this report. We would like to express our special appreciation and thanks to our advisor Dr. Muhammad Akram, who have been a tremendous mentor for us. We are thankful for his aspiring guidance, invaluably constructive criticism and friendly advice during the project work. We are sincerely grateful to him for sharing his truthful and illuminating views on a number of issues related to the project.

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A special thanks to our families. Words cannot express how grateful we are to our parents for all of the sacrifices that they made on our behalf. Their prayers for us were what sustained us thus far. We would also like to thank all of my friends who supported us in writing, and incented us to strive towards our goal.

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Chapter 1

1 INTRODUCTION

1.1 Defibrillator

A defibrillator is a device that introduces electrical current or shock to the heart to correct cases of ventricular fibrillation (a quivering of the heart which limits the heart muscle from pumping blood to the body), cardiac arrhythmias (abnormal electrical activity), and ventricular tachycardia (fast heart rhythm). It acts on the heart’s natural pacemaker, the sinoatrial node. Without the aid of defibrillation, or the process of using a defibrillator, these abnormalities encountered can lead to more serious complications or even death.

This is usually a heavy device that used in hospitals and required a medical staff to operate it and time is needed to take the patient to the hospital hence risk of life in it.

1.2 External Defibrillator Vest

The external defibrillator vest is a treatment option for sudden cardiac arrest that offers patients advanced protection and monitoring as well as improved quality of life.

The EDV is a wearable defibrillator. Unlike an implantable cardioverter defibrillator (ICD), the EDV is worn outside the body rather than implanted in the chest. This device continuously monitors the patient's heart with dry, non-adhesive sensing electrodes to detect life-threatening abnormal heart rhythms. If a life-threatening rhythm is detected, the device alerts the patient prior to delivering a treatment shock, and thus allows a conscious patient to delay the treatment shock.

The EDV is lightweight and easy to wear, allowing patients to return to their activities of daily life, while having the peace of mind that they are protected from sudden cardiac arrest. The EDV is non-invasive and consists of two main components, a garment and a monitor. The garment, worn under the clothing, detects arrhythmias and delivers treatment shocks. The monitor is worn around the waist or from a shoulder strap and continuously monitors the patient’s heart.

Table 1-1 External Defibrillator Vs External Defibrillator Vest

External Defibrillator	External Defibrillator Vest
External Defibrillator is used in hospital	External Defibrillator Vest can be used by every sensible person easily
External Defibrillator is operated by medical staff	External Defibrillator Vest is automatically operated
External Defibrillator takes time to proceed its process	External Defibrillator Vest delivers shock after few seconds of heart attack
External Defibrillator is heavy apparatus	External Defibrillator Vest is light weight and easy to carry

1.3 External Defibrillator Vest Layout Proposed

The proposed External Defibrillator Vest consists of four major components in which first of all ECG Simulator that produced the ECG then we use Arduino Mega that continuously monitor and observe the ECG and in the event of abnormal ECG an alarm beeps and warns the patient that shock is about to deliver meanwhile capacitor is charged and after few seconds defibrillator produce shock to the patient through electrodes.

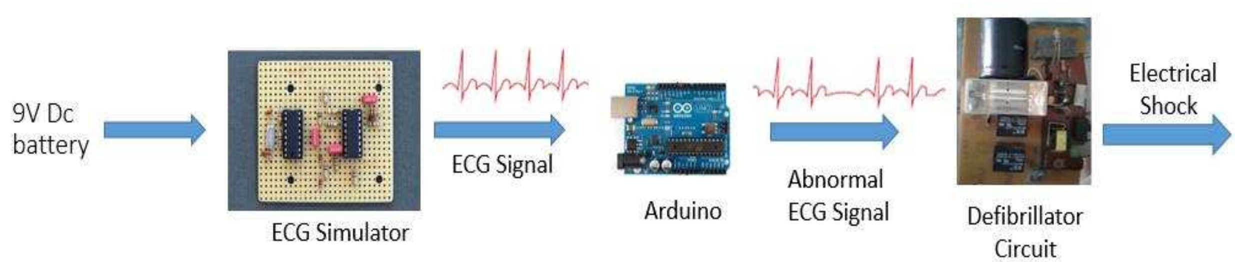

Figure 1-1 Layout of External Defibrillator Vest

1.4 Problem Statement

Death rate due to fibrillation and sudden heart attack have become the largest cause of natural death. In order to reduce the death rate the patient should be treated as soon as possible to increase the chances of resuscitation. EDV is used for this purpose to continuously monitor the patient's heart activity to detect life-threatening abnormal heart rhythms.

To provide the patients with something that is comfortable and easy to wear and also feasible to buy at the same time, there should be an EDV that is light weight and cost-effective i.e easily accessible to people belonging to all classes of society. These two desired attributes should be combined in a single product so that people will benefit themselves from these kind of vests as much as they can.

1.5 Aims & Objectives

The objectives of this project are to design a light weighted and low cost external defibrillator vest to continuously monitor the patient's heart to detect life-threatening abnormal heart rhythms. If a life-threatening rhythm is detected, the device alerts the patient prior to delivering a treatment shock, and thus allows a conscious patient to delay the treatment shock.

1.6 Scope of the Project

In conducting a project, the scopes of the project must be realistic in order to ensure that the project is done within the time with the objectives reached.

1. The EDV must be light weighted.
2. Analog to Digital Conversion of ECG signals.
3. Continuous measurement of ECG from a person who will wear EDV
4. Control method of EDV to detect the QRS peak of ECG.
5. Shocking Circuit that will operate for that person whose heart rhythm become abnormal.

1.7 Thesis Structure

This thesis discusses about the design which is include the hardware design, software and circuit design of the External Defibrillator Vest (EDV). First chapter will discussed about the introduction of the project. Problem statements, objectives of the project and the scope of work are discussed in this chapter. In second chapter, the literature review of the project is discussed. In chapter 3, the hardware design, software and circuit design, and the programming of the EDV are discussed as the research methodology. Results and discussion will be discussed in Chapter 4. Finally chapter 5 will is about the conclusion and future recommendation.

Chapter 2

2 LITERATURE REVIEW

External defibrillator vest project require extensive research and knowledge regarding Defibrillation. Defibrillation is a treatment for life-threatening cardiac dysrhythmias, specifically ventricular fibrillation (VF) and non-perfusing ventricular tachycardia (VT). A defibrillator delivers a dose of electric current (often called a counter shock) to the heart. The electrical shock does not have to be timed with the heart's intrinsic cardiac cycle. This depolarizes a large amount of the heart muscle, ending the dysrhythmia. Subsequently, the body's natural pacemaker in the sinoatrial node of the heart is able to re-establish normal sinus rhythm .In all these procedures require a specific time and in the case of late defibrillation a person may lose his life. So a best procedure of this external defibrillation to use it as a vest so that every second its monitoring the heart beat and in the case of abnormal rhythm its provides instantaneously shock to the patient to restore its heart beat.

The outcomes of patients with cardiac arrests in-hospital are still unacceptably poor. The prospective observational study from a multicenter registry (NRCPR) of cardiac arrests in 253

US and Canadian hospitals shows that the survival is 36% for patients with ventricular fibrillation and pulseless ventricular tachycardia and 11% for patients with pulseless electrical activity [1]. Cardiac arrests with shockable rhythm in the operating rooms are associated with survival rates between 39.6 -46.8% [2]. Survival rates depend on the time to defibrillation and early, effective and continued chest compressions [3].

Although AEDs are not frequently used in the operating rooms, the use of AED in hospital wards, has increased significantly between 2003 and 2008. This is the result of studies demonstrating a 1.75-fold improved survival of out-of-hospital cardiac arrest patients [4] with use of an AED. Until the publication of Chan’s study, only limited data was available regarding the success rates of automated external defibrillators. The data were conflicting and inconclusive in large part of because the studies were underpowered.

Continued efforts to improve survival and outcomes of cardiac arrest victims of in-hospital patients and a push from manufacturers has resulted in increased AED use in many hospitals. Advocates of AED use suggested that the majority of first responders in the hospital were not trained in recognizing cardiac rhythms and that the device would allow for early recognition of shockable rhythms and thus improved survival. The study by Forcina et al, was the first to indicate that in-hospital use of AED may not be superior to manual external defibrillators for patient survival. The outcomes of patients with shockable rhythms were the same and the survival of patients who had non-shockable rhythm was worse (15% with AED vs. 15% without AED; p=0.04) [5]. This data is now supported by Chan's study which demonstrated similar results on a large number of patients [6].

As a member of the cardiac arrest response teams, anesthesiologists should be aware of the type of the defibrillator devices used in the hospital and the modes in which they are used. Aggressive chest compressions with minimal interruptions should be emphasized in any code situation and the use of defibrillators in a manual mode, as soon as a ACLS trained personnel is available, should be promoted.

2.1 Main Components

In this section, a theoretical overview id provided for some of the components used in a External Defibrillator Vest.

2.1.1 Electrodes

The electrodes are the components through which the defibrillator delivers energy to the patient's heart. Many types of electrodes are available including hand-held paddles, internal paddles, and self-adhesive, pre-gelled disposable electrodes. In general, disposable electrodes are preferred in emergency settings because they have advantages such as increasing the speed of shock and improving defibrillation technique. The paddle size affects the current flow. Larger paddles create a lower resistance and allow more current to reach the heart. Thus, larger paddles are more desirable. Most manufacturers offer adult paddles, which are between 3.1-5.1 in (8-13 cm) in diameter, and pediatric paddles, which are smaller.

Figure 2-1 Sensor Cable

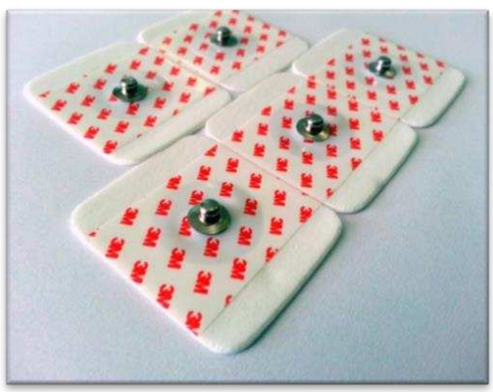

Figure 2-2 ECG Electrodes

Since skin is a poor conductor of electricity, a gel must be used between the electrode and the patient. Without this conductor, the level of the current reaching the heart would be reduced. Also, the skin may be burned. A variety of gels and pastes are available for this purpose. These are composed of cosmetic ingredients like lanolin or petrolatum. Chloride ions in the formula also help form a conductive bridge between the skin and the electrode allowing better charge transfer. Many of these materials are the same compounds used for other medical devices such as ECG scans.

2.1.2 Battery

Batteries are essentially containers of chemical reactions. In defibrillators, a variety of batteries are used. They are characterized by the chemical reactions contained in them and include lead-acid, lithium, and nickel-cadmium systems. These batteries can typically be recharged by an outside power source, and when not in use defibrillators are stored plugged in. Since extreme temperatures negatively affect the batteries, defibrillators are stored in controlled environments. Over time batteries wear out and are replaced. This is important because battery chemistries are inherently corrosive and potentially toxic. We use Dura Cell 9V alkaline battery in our project.

Figure 2-3 Alkaline Battery

2.1.3 Capacitor

The most important component of a defibrillator is a capacitor that stores a large amount of energy in the form of electrical charge, then releases it over a short period of time. A capacitor consists of a pair of conductors (e.g. metal plates) separated by an insulator (called a dielectric).

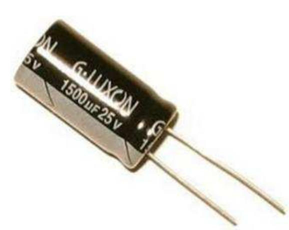

Figure 2-4 Charging Capacitor

2.1.4 Transformer

A transformer is an electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. A varying current in one coil of the transformer produces a varying magnetic field, which in turn induces a voltage in a second coil. Power can be transferred between the two coils through the magnetic field, without a metallic connection between the two circuits. We use step up transformer in our project because we used high voltage for defibrillation.

Figure 2-5 Auto Transformer

2.1.5 Transistor

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. It is composed of semiconductor material usually with at least three terminals for connection to an external circuit.

Figure 2-6 Transistor

2.1.6 Trigger Capacitor

This Capacitor is used for trigger coils, where the lamps are relatively small and the wires to the lamp are short.

Figure 2-7 Trigger Capacitor

2.1.7 Neon Lamp

A neon lamp (also neon glow lamp) is a miniature gas discharge lamp. The lamp typically consists of a small glass capsule that contains a mixture of neon and other gases at a low pressure and two electrodes (an anode and a cathode). When sufficient voltage is applied and sufficient current is supplied between the electrodes, the lamp produces an orange glow discharge. The glowing portion in the lamp is a thin region near the cathode; the larger and much longer neon signs are also glow discharges, but they use the positive column which is not present in the ordinary neon lamp. Neon glow lamps are widely used as indicator lamps in the displays of electronic instruments and appliances. We use neon lamp in our project to indicate that capacitor is fully charged.

Figure 2-8 Neon Lamp

2.1.8 Trigger Coil

Once the main storage capacitor has finished charging, a smaller capacitor is discharged into the trigger transformers primary coil. A large magnetic pulse is generated inside the transformer, which is coupled to the secondary coil through the core.

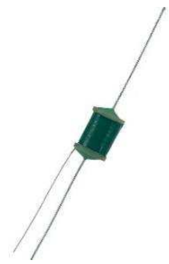

Figure 2-9 Trigger Coil

2.1.9 Relay

A relay is an electrically operated switch. Many relays use an electromagnet to mechanically operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a separate low-power signal, or where several circuits must be controlled by one signal. We use relay in our project to discharge the capacitor.

Figure 2-10 Relay

2.1.10 Flash Light

We used flash light in our project to the indication that capacitor discharged and electric shock deliver to the patient.

Figure 2-11 Flash Camera

2.1.11 Controller

We use Arduino Mega in our project as a controller that observe the ECG and in the event of abnormal ECG its gives signal to defibrillator circuit to delivers the shock to the patient.

Figure 2-12 Arduino Mega

2.2 Historical Background

An external defibrillator is a device that delivers an electric shock to the heart through the chest wall. This shock helps restore the heart to a regular, healthy rhythm. The device is generally sold as a kit that consists of a power control unit, paddle electrodes, and various accessories. The parts are made individually and pieced together via an integrated production process. Since then medical device manufacturers have introduced various defibrillators, internal and external, that have added years to patients' lives.

Figure 2-13 Old Defibrillator

2.2.1 Working of heart

To understand how a defibrillator can restart a stalled heart, the physiology of the organ must be considered. The human heart has four chambers, which create two pumps. The right pump receives the oxygen-depleted blood returning from the body and pumps it to the lungs. The left pump receives the oxygenated blood from the lungs and pumps it to the rest of the body. Both pumps have a ventricle chamber and an atrium chamber and operate in a similar manner. The blood collects in the atrium and is then transferred to the ventricle. Upon contraction, the ventricle pumps the blood away from the heart.

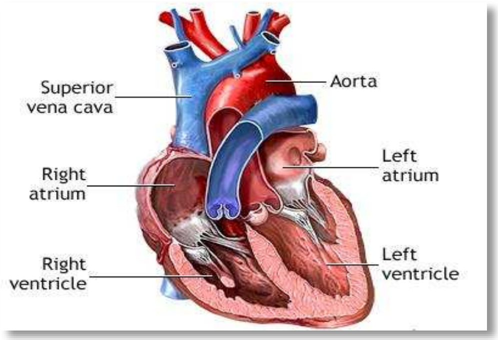

Figure 2-14 Internal Structure of Heart

The coordination of the pumping action is critical for the heart to function correctly. A pacemaker region, which is located in the heart's right atrium, is responsible for this control. In this region, a spontaneous electrical impulse is created by the diffusion of calcium ions, sodium ions, and potassium ions across the cell membranes. The impulse thus created is transferred to the atrium chambers causing them to contract, pushing blood into the ventricles. After about 150 milliseconds the impulse moves to the ventricles, which causes them to contract and pump blood out of the heart. As the impulse moves away from the chambers of the heart, these sections relax. In a normal heart, the process then repeats itself.

In some cases, the electrical control system of the heart malfunctions and results in an irregular heartbeat such as ventricular fibrillation. Various conditions can cause ventricular fibrillation including blocked arteries, poor reaction to anesthesia, and electrical shock. Defibrillators are used to supply a strong electrical shock to the heart. Two electrodes are placed on the chest and a shock is given. A typical defibrillator device will deliver a shock for three to nine milliseconds. For reasons not quite understood, the shock essentially resets the natural ventricular rhythm and allows the heart to beat normally.

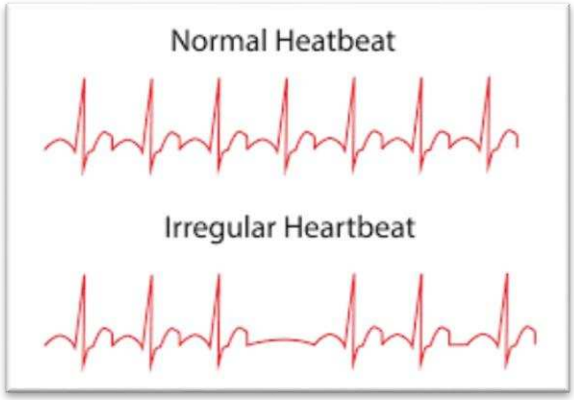

Figure 2-15 Normal Vs A abnormal Rhythms of Heart

In practice, an external defibrillator can be operated at an emergency site or a hospital. The operator first turns on the machine and then applies a conductive gel to the paddle electrodes or patient's chest. The energy level is selected and the instrument is charged. The paddles are placed firmly on the patient's unclothed chest with a pressure of about 25 lb (11 kg). The buttons on the electrodes are pressed simultaneously and the electric shock is delivered. The patient is then monitored for a regular eat. The process is repeated if necessary.[7]

2.2.2 History

The discovery that a misfiring heart could be restarted using an electrical charge is one of the great developments of modern medicine. This idea was begun around 1888 when it was suggested by Mac William that ventricular fibrillation might be the cause of sudden death. Ventricular fibrillation is a condition in which the heart suddenly beats irregularly, preventing its blood-pumping ability that ultimately can lead to death. It can be caused by a coronary artery blockage, various anesthesia, and electric shock.[8]

During the 1920s and 1930s, research in this field was supported by the power companies because electric shock induced ventricular fibrillation killed many power utility line workers. Hooker, William B. Kouwenhoven, and Orthello Langworthy produced one of the first successes of this research. In 1933, they published the results of an experiment, which demonstrated that an internally applied alternating current could be used to produce a counter shock that reversed ventricle fibrillation in dogs.

In 1947, Dr. Claude Beck reported the first successful human defibrillation. During a surgery, Beck saw his patient experiencing a ventricular fibrillation. He applied a 60 Hz alternating current and was able to stabilize the heartbeat. The patient lived and the defibrillator was born. In 1954, Kouwenhoven and William Milnor demonstrated the first closed chest defibrillation on a dog. This work involved the application of electrodes to the chest wall to deliver the necessary electric counter shock. In 1956, Paul Zoll used the ideas learned from Kouwenhoven and performed the first successful external defibrillation of a human.

William Bennett Kouwenhoven was born January 13, 1886 in Brooklyn. Trained as an electrical engineer, his most enduring contributions to science came from the medical arena. Using his electrical engineering background, Kouwenhoven invented three different defibrillators and developed cardiopulmonary resuscitation (CPR) techniques.[9]

In the 1920s, Kouwenhoven's interest crossed between electrical engineering and medicine. His engineering work focused on high tension wire transmission of electricity. Kouwenhoven became interested in electricity's possible role in reviving animals. He knew that when applied to the heart an electric current could start it again.

From 1928 through the mid-1950s, Kouwenhoven developed three defibrillators: the open-chest defibrillator, the Hopkins AC Defibrillator, and then Mine Safety Portable. These were intended for use within two minutes of the start of ventricular fibrillation, and at least one required direct contact with the heart. In 1956, Kouwenhoven began developing a non-invasive method. During an experiment on a dog, he realized the weight of the defibrillator's paddles raised the animal's blood pressure. Based on this Kouwenhoven developed CPR.

In the 1960s, scientists discovered that direct current defibrillators had fewer adverse side effects and were more effective than alternating current defibrillators. In 1967, Pantridge and Geddes demonstrated that using a mobile, battery-powered DC defibrillator could save lives. The late sixties saw the introduction of an implantable defibrillator by Dr. Michael Mirowski. Both internal and external defibrillators were redesigned in the 1970s to automatically detect ventricular fibrillation. As improvements in electronics and computers became available these technologies were adapted to defibrillators.[10]

Today, defibrillation has become an integral part of the emergency response routine. In fact, the American Heart Association considers defibrillation a basic life support skill for paramedics and rescue workers.

Figure 2-16 Modern Defibrillator

Chapter 3

3 METHDOLOGY

3.1 Design

This project is divided into four parts which are ECG Simulator, Defibrillator Circuit Design, audio output and algorithm part. In this chapter, the methodology of the project which is consisting of the four parts that mentioned before will be discussed

3.2 ECG Simulator

3.2.1 Why we need ECG Simulator?

An ECG (electrocardiogram) simulator is an electronic tool that plays an essential role in the testing, design, and development of ECG monitors and other ECG equipment. Principally and ECG simulator provides ECG monitors with an electrical signal that emulates the human heart's electrical signal so that the monitor can be tested for reliability and important diagnostic capabilities. However, the current portable commercially available ECG simulators are lacking in their ability to fully test ECG monitors. Specifically, the portable simulators presently on the market do not produce authentic ECG signals but rather they endeavor to create the ECG signals mathematically. They even attempt to mathematically create arrhythmias (irregular heartbeats on which there are many different types). Arrhythmia detection is an important capability for any modern ECG monitor because arrhythmias are often the critical link to the diagnosis of heart conditions or cardiovascular disease.[11]

Figure 3-1 Serial monitor for ECG Simulator

Testing of simulator was done using simulation Arduino. ECG values were fed and beats per minute (bpm) were calculated and displayed on Arduino as shown in Figure 3-1. When bpm values are lower than 60 and greater than 100, then our relay operate and defibrillator produces an electrical shock. This defibrillator is designed considering the normal heart rate which is 60 to 100. People with good physical fitness, e.g. athletes generally have heart rates below 60, for them the threshold can be adjusted.

3.3 Audio output from Arduino

We need to alarm the person that shock is deliver to him. For this purpose, we need some sound that aware it from the electric shock. To produce sound we need the following hardware to design the system

3.3.1 Hardware

- Arduino
- SD Card Adopter
- SD Card
- Amplifier
- Speaker

Figure 3-2 SD Card Adaptor

Table 3-1 Interfacing of SD Card Adaptor with Arduino

SD Card Adopter	Arduino
Vcc	5V
MISO	Digital Pin 12
MOSI	Digital Pin 11
SCK	Digital Pin 13
CS	Digital Pin 10
Ground	Ground

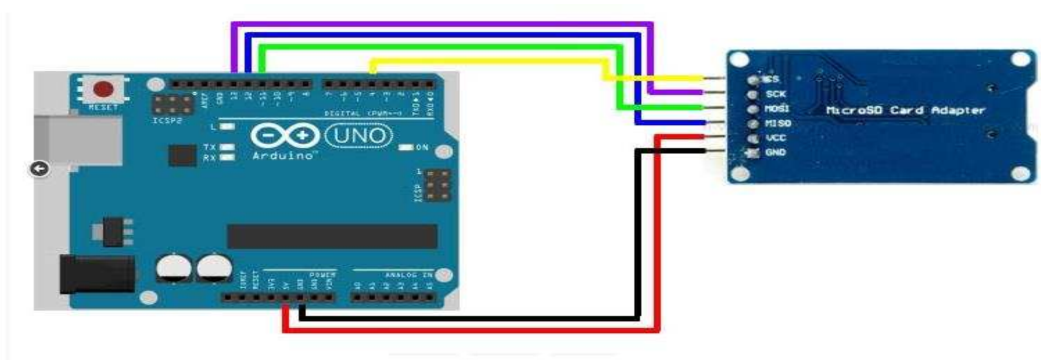

Figure 3-3 Connections of Arduino with the SD Card Adaptor

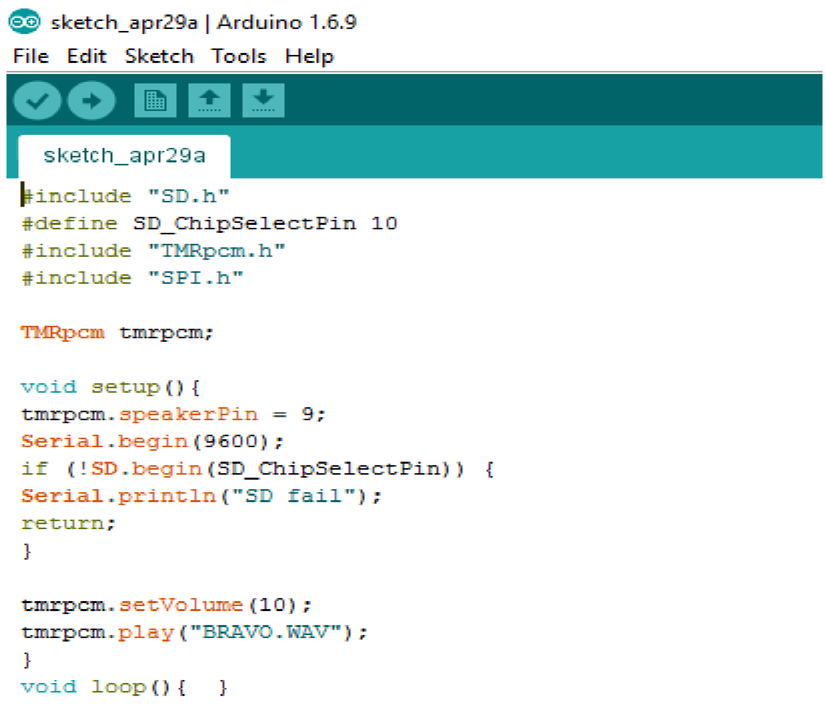

Figure 3-4 Audio Output from Arduino Program

3.3.2 Audio Amplifier

An audio amplifier is an electronic device that increases the strength (amplitude) of audio signals that pass through it. An audio amplifier amplifies low-power audio signals to a level which is suitable for driving loudspeakers.

Components we used in our audio amplifier are listed below

- 470 µF Capacitor
- 0.1 µF Capacitor
- 2.2 µF Capacitor
- 1000 µF Capacitor
- TDA02002 Transistor
- 100 K Variable resister
- 1N4007 Diode

Figure 3-5 Audio Amplifier

3.4 Software Design

3.4.1 Algorithm

Algorithm implemented to process ECG signal and to detect QRS-complexes is Pan+Tompkins algorithm. The algorithm is able to correctly detect QRS complexes in the presence of the severe noise typical of the ambulatory ECG environment. The algorithm uses three different types of processing steps: linear digital filtering, nonlinear transformation, and decision rule algorithms. First, in order to attenuate noise, the signal passes through a digital bandpass filter composed of cascaded high-pass and lowpass filters. The next process after filtering is differentiation and then moving window integration. Information about the slope of the QRS is obtained in the derivative stage. The squaring process intensifies the slope of the frequency response curve of the derivative and helps restrict false positives caused by T waves with higher than usual spectral energies. The moving window integrator produces a signal that includes information about both the slope and the width of the QRS complex. The algorithm is divided into three processes: learning phase1, learning phase 2, and detection. Learning phase 1 requires about 2 s to initialize detection thresholds based upon signal and noise peaks detected during the learning process. Learning phase 2 requires two heartbeats to initialize RR -interval average and RR interval limit values. The subsequent detection phase does the recognition process and produces a pulse for each QRS complex. The thresholds and other parameters of the algorithm are adjusted periodically to adapt to changing characteristics of the signal.[12]

Two sets of thresholds are used to detect QRS complexes. One set thresholds the filtered ECG, and the other thresholds the signal produced by moving window integration. By using thresholds on both signals, the reliability of detection is improved compared to using one waveform alone. Pre-processing the ECG with this digital bandpass filter improves the signal-to noise ratio and permits the use of lower thresholds than would be possible on the unfiltered ECG. This increases the overall detection sensitivity. The detection thresholds float over the noise that is sensed by the algorithm. This approach reduces the number of false positives caused by types of noise that mimic the characteristics of the QRS complex [13]. The algorithm uses a dual-threshold technique to find missed beats, and thereby reduce false negatives. There are two separate threshold levels in each of the two sets of thresholds. One level is half of the other. The thresholds continuously adapt to the characteristics of the signal since they are based upon the most-recent signal and noise peaks that are detected in the ongoing processed signals. If the program does not find a QRS complex in the time interval corresponding to 166 percent of the current average RR interval, the maximal peak detected in that time interval that lies between these two thresholds is considered to be a possible QRS complex, and the lower of the two thresholds is applied.

For the case of irregular heart rates, both thresholds are reduced by half in order to increase the sensitivity of detection and to avoid missing valid beats. Once a valid QRS complex is recognized, there is a 200 ms refractory period before the next one can be detected since QRS complexes cannot occur more closely than this physiologically. This refractory period eliminates the possiblity of a false detection such as multiple triggering on the same QRS complex during this time interval. When a QRS detection occurs following the end of the refractory period but within 360 ms of the previous complex, we must determine if it is a valid QRS complex or a T wave. In this case, we judge the waveform with the largest slope to be the QRS complex. [14]

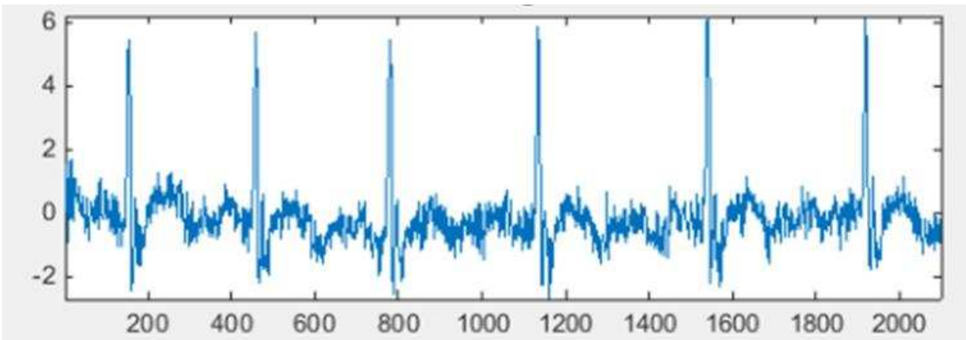

Figure 3-6 Raw signal of ECG

3.4.2 Bandpass Filter

The desirable passband to maximize the QRS energy is approximately 5-15 Hz .A low pass and high pass filters are used as a cascade to approximately implement passband.

3.4.3 Low-Pass Filter

The transfer function of the second-order low-pass filter is

$$H(z) = \frac{(1 - z^{-6})^2}{(1 - z^{-1})^2} \tag{3.1}$$

The amplitude response is

$$|H(wT)| = \frac{Sin^2(3wT)}{Sin^2(\frac{wT}{2})} \tag{3.2}$$

Where T is the sampling period. The difference equation of the filter is

$$y(nT) = 2y(nT - 2T) - y(nT - 2T) + x(nT) - 2x(nT - 6T) + x(nT - 12T) \tag{3.3}$$

The cutoff frequency of the filter is 12 Hz.

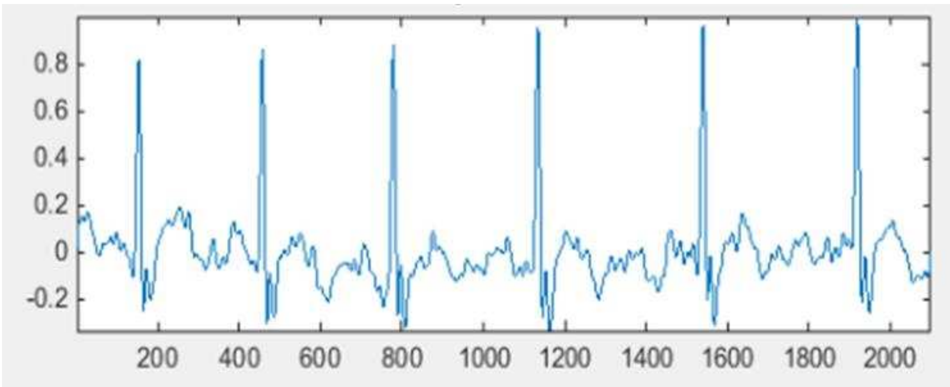

Figure 3-7 ECG signal after passing by Low pass filter

3.4.4 High-Pass Filter

The transfer function of the high-pass filter is

$$H(z) = \frac{(-1 + 32z^{-16} + z^{-32})}{(1 + z^{-1})} \tag{3.4}$$

The amplitude response is

$$|H(wT)| = \frac{[256 + sin^2\ (16wT)]^{\frac{1}{2}}}{\cos(\frac{wT}{2})} \tag{3.5}$$

The difference equation is

$$y(nT) = 32x(nT - 16T) - [y(nT - T) + x(nT) - x(nT - 32T)] \tag{3.6}$$

The cut-off frequency of filter is 5 Hz.

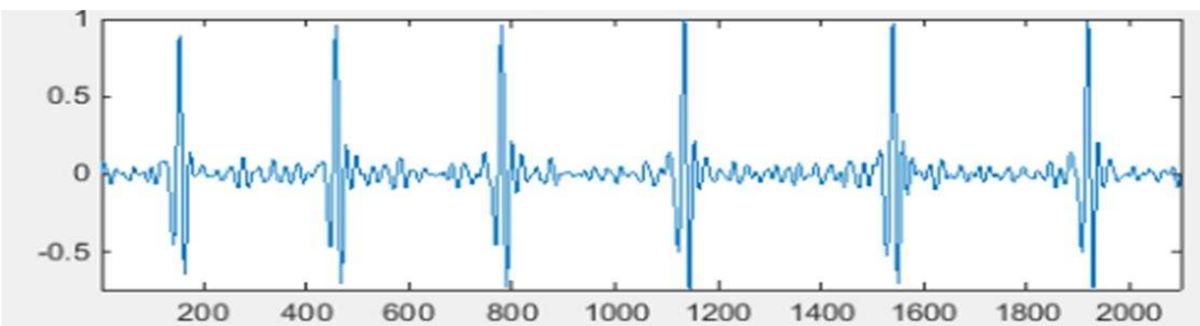

Figure 3-8 ECG signal after passing by High pass filter

3.4.5 Derivative

After filtering, the signal is differentiated to provide the QRS complex slope information with derivative of transfer function

$$H(z) = \frac{\frac{1}{8}T}{(-z^{-2} - 2z^{-1} + 2z^{1} + z^{2})} \tag{3.7}$$

The amplitude response is

$$|H(wT)| = \left(\frac{1}{4}T\right)[Sin(2wT) + 2Sin(wT)] \tag{3.8}$$

The difference equation is

$$y(nT) = \left(\frac{1}{8}T\right)[-x(nT-2T) - 2x(nT-T) + 2x(nT+T) + x(nT+2T)] \tag{3.9}$$

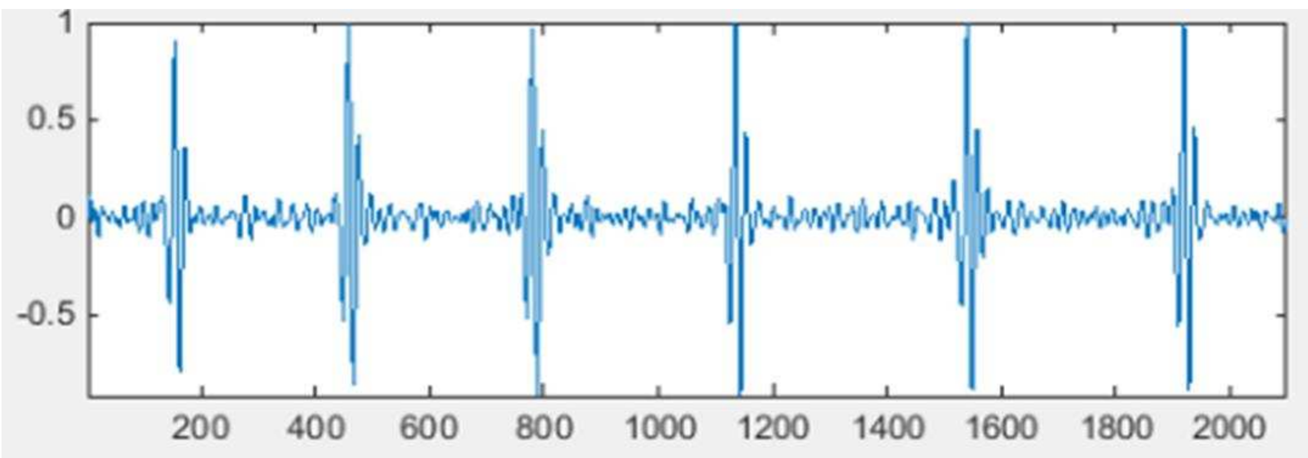

Figure 3-9 ECG signal after passing by High pass filter

3.4.6 Squaring

After differentiation, the signal is squared point by point. The equation of this operation is

$$y(nT) = [x(nT)]^2 \tag{3.10}$$

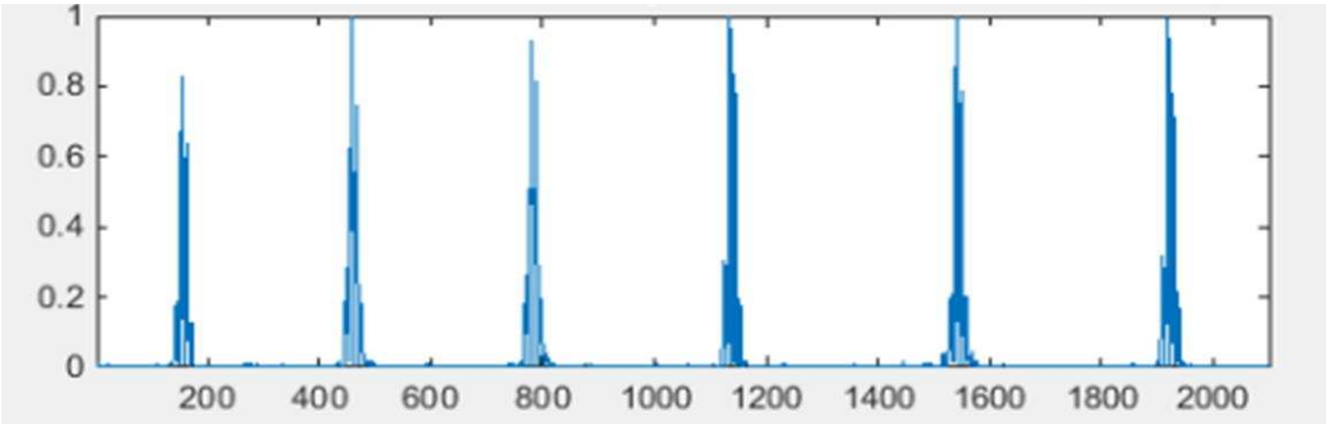

Figure 3-10 ECG signal after passing by squared filter

3.4.7 Integration

The purpose of moving-window integration is to obtain waveform feature information in addition to the slope of the R wave. The width of the window is determined empirically. For our sample rate of 200 samples/s, the window is 30 samples wide (150 ms).

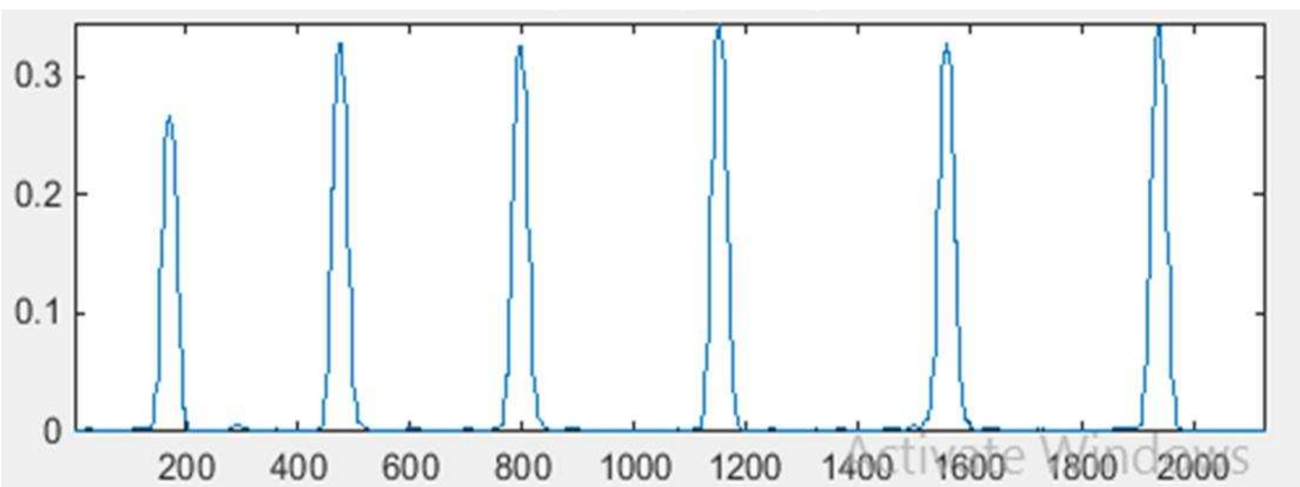

Figure 3-11 ECG signal after passing integration filter

3.4.8 Thresholds

The set of thresholds initially applied to the integration waveform is computed from

SPKI = 0.125 PEAKI + 0.875 SPKI (3.10)

(if PEAKI is the signal peak)

NPKI = 0.125 PEAKI + 0.875 NPKI (3.11)

(if PEAKI is the noise peak)

THRESHOLD Il = NPKI + 0.25 (SPKI - NPKI) (3.12)

THRESHOLD I2 = 0.5 THRESHOLD Il (3.13)

Where all the variables refer to the integration waveform:

PEAKI is the overall peak,

SPKI is the running estimate of the signal peak,

NPKI is the running estimate of the noise peak,

THRESHOLD Il is the first threshold applied, and

THRESHOLD 12 is the second threshold applied.

The set of thresholds applied to the filtered ECG is determined from

SPKF = 0.125 PEAKF + 0.875 SPKF (3.14)

(if PEAKF is the signal peak)

NPKF = 0.125 PEAKF + 0.875 NPKF (3.15)

(if PEAKF is the noise peak)

THRESHOLD F1 = NPKF + 0.25 (SPKF - NPKF) (3.16)

THRESHOLD F2 = 0.5 THRESHOLD Fl (3.17)

Where all the variables refer to the filtered ECG:

PEAKF is the overall peak,

SPKF is the running estimate of the signal peak,

NPKF is the running estimate of the noise peak,

THRESHOLD Fl is the first threshold applied, and

THRESHOLD F2 is the second threshold applied.

To be identified as a QRS complex, a peak must be recognized as such a complex in both the integration and bandpass-filtered waveforms.

$$RR\ AVERAGE2 = 0.125(RR'_{n-6} + \cdots + RR'_{n}) \qquad (3.18)$$

Where RR'_n is the most recent RR interval that fell between the acceptable low and high RR interval limits. The RR-interval limits are

RR LOW LIMIT = 92% RR AVERAGE2 (3.19)

RR HIGH LIMIT = 116% RR AVERAGE2. (3.20)

RR MISSED LIMIT = 166% RR AVERAGE2 (3.21)

If a QRS complex is not found during the interval specified by the RR MISSED LIMIT, the maximal peak reserved between the -two established thresholds is considered to be a QRS Candidate.

3.4.9 T-wave Identification

When an RR interval is less than 360 ms (it must be greater than the 200 ms latency), a judgment is made to determine whether the current QRS complex has been correctly identified or whether it is really a T wave. If the maximal slope that occurs during this waveform is less than half that of the QRS waveform that preceded it, it is identified to be a Twave; otherwise, it is called a QRS complex.

3.4.10 Software Modeling

The above mentioned algorithm is implemented using MATLAB. The MATLAB model for QRS detection and heart rate determination is presented below:

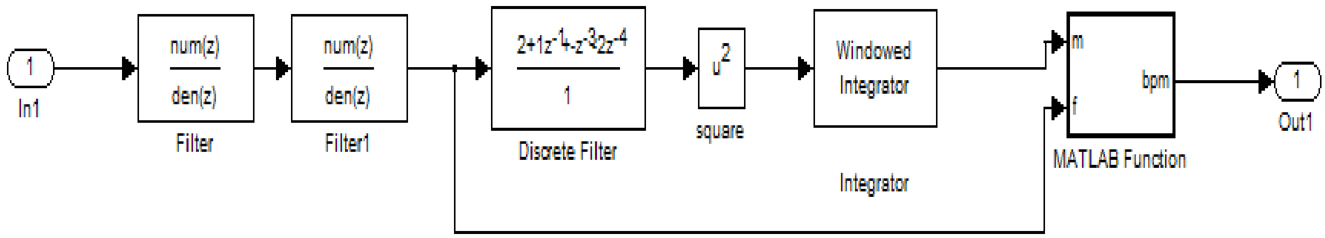

Figure 3-12 QRS model of MATLAB

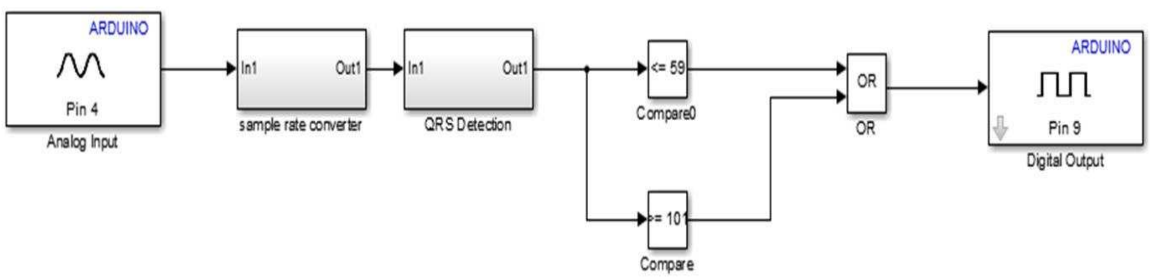

Figure 3-13 Modeling on MATLAB

3.5 Defibrillator Circuit

Keeping our goals in mind, defibrillator circuit is designed using high frequency audio transformer. An alkaline battery of 9Vand 800-900mA is employed and are stepped to 1000Volts using transformer of 1200 to 8 ohms audio transformer fed by a transistor D826 (oscillator).A neon lamp is used to indicate the status of capacitor voltages. When neon lamp on, it indicated about the charging of the capacitor and almost on 800-850V neon lamp shows indication. The capacitor used to store charge is electrolytic capacitor that can be stored till a relay signal is fired to discharge the capacitor. Since there is no provided part for the discharging of capacitor, it stored charge in it. Relay is controlled by Arduino board and depending upon the system being monitored. When the abnormal ECG indicated, capacitor going to discharge through flash light. Because our project is a prototype and can't implemented on a patient that's why we use flash light to indicate that our capacitor is discharged and neon lamp is also off when capacitor discharged it is another indication of discharging.

Here is the proposed Layout for the Defibrillator circuit

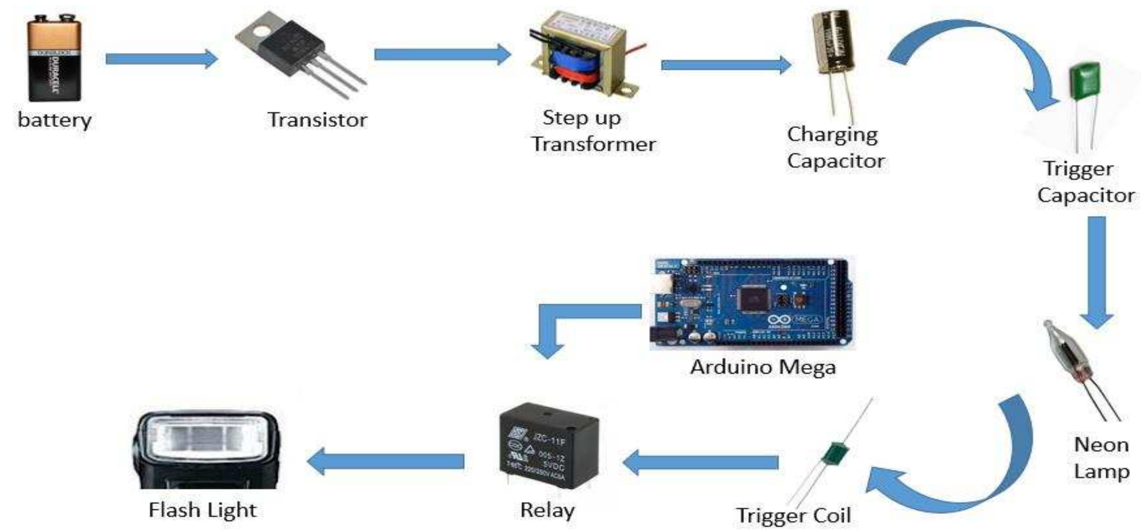

Figure 3-14 Layout of Defibrillator Circuit

3.5.1 Calculations of Power

A defibrillator is a device that applies a strong electric shock to the chest over a time of a few milliseconds. The device contains a capacitor of several microfarads, charged to several thousand volts.

The capacitor we used in our project is 20.0 μF and is charged up to 1000 volts

Here are some important equations that we used in our calculations [16].

$$Capacitance = C = \frac{Q}{V} \tag{3.22}$$

$$Energy\ Stored = \frac{1}{2}CV^2 \tag{3.23}$$

$$Current = I = \frac{Q}{t} \tag{3.24}$$

$$Power = P = VI \tag{3.25}$$

How much charge is stored in the capacitor before it is discharged?

$$Q = CV = 20e^{-6} * 1000 = 0.02\ C$$

How much energy is released when the capacitor is discharged?

$$Energy\ Stored = \frac{1}{2}CV^2 = \frac{1}{2}20e^{-6} * 1000^2 = 10J$$

If the capacitor completely discharges in 2.0 ms, what is the average current delivered by the defibrillator?

$$I = \frac{Q}{t} = \frac{0.02}{0.02} = 1A$$

What is the average power delivered?

$$P = VI = 1000 * 1 = 1000\ W$$

Figure 3-15 Hardware of Defibrillator

Chapter 4

4 TESTING AND RESULTS

4.1 **Measuring ECG from a Patient Body**

An ECG Sensor with disposable electrodes attaches directly to the chest to detect ecg.. ECG Sensors is very light weight, slim and accurately to measures continuous heart beat and give rate data of heart beat.

Electrodes of ECG Sensor have 3 pins and connected by cable which is 30 inches in length. It makes ECG sensor easy to connect with controller and placed at the waist or pocket. In additional, the plug-in for the cable is a male sound plug which will make the cable to easily removed or inserted into the amplifier board. The sensor assembled on an arm pulse and a leg pulse. All of every sensor electrodes have methods to assemble in body.

4.2 Hardware and Software Requirements:-

4.2.1 Hardware

1) Arduino Uno/Mega/Nano

2) ECG Module (AD8232)

3) ECG Electrodes - 3 pieces

4) ECG Electrode Connector -3.5 mm

5) Oscilloscope

6) Connecting Wires

4.2.2 Software Requirement

1) Arduino IDE

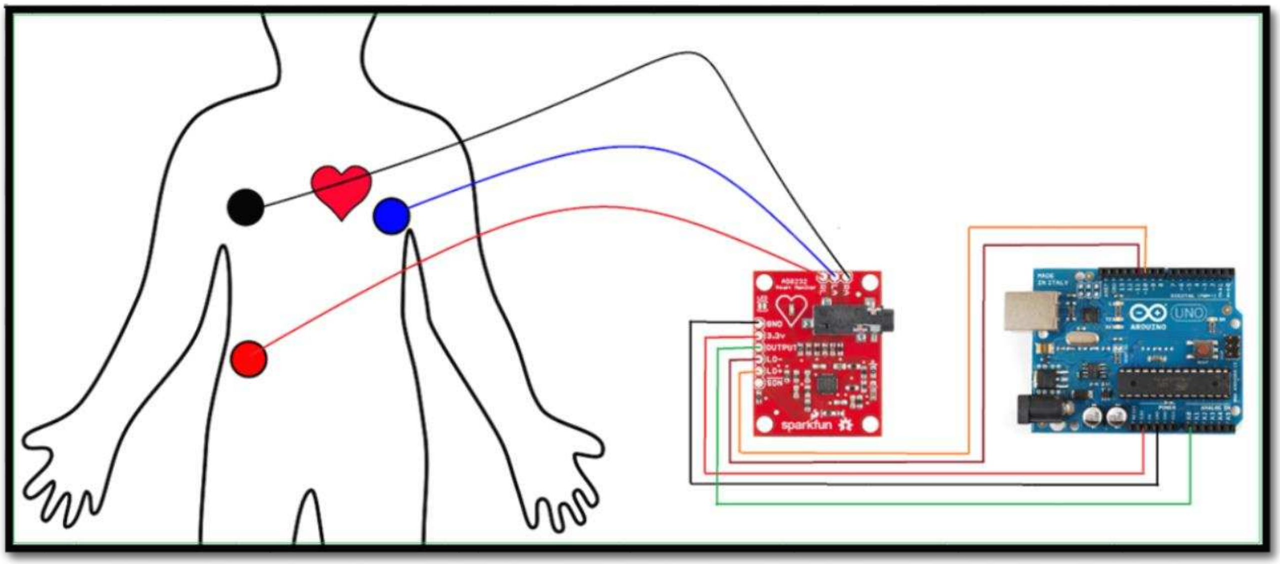

Figure 4-1 Interfacing diagram of Arduino and AD8232

Table 4-1 Connections of AD8232 with Arduino

Arduino	AD8232
Arduino 3.3V	3.3V pin
Arduino pin 10	L0+
Arduino Pin 11	L0-
Arduino Analog 1 (A1)	Output
Arduino Gnd	Gnd

4.2.3 ECG Module (AD8232) Specifications

The AD8232 module breaks out nine connections from the IC that you can solder pins, wires, or other connectors to. SDN, LO+, LO-, OUTPUT, 3.3V, GND provide essential pins for operating this monitor with an Arduino or other development board. Also provided on this board are RA (Right Arm), LA (Left Arm), and RL (Right Leg) pins to attach our own custom sensors. Additionally, there is an LED indicator light that will pulsate to the rhythm of a heartbeat.

4.2.3 Features of ECG Module

- Operating Voltage - 3.3V
- Analog Output
- Leads-Off Detection
- Shutdown Pin
- LED Indicator
- 3.5mm Jack for Biomedical Pad Connection or Use 3 pin header.

Figure 4-2 ECG Module (AD8232)

4.2.4 ECG Serial Monitoring

In Arduino Serial monitoring we can check that our all system is working properly or not. If we have some output it means that our pulses are truly observed by the Arduino serial monitor.

Here is the program of ECG serial monitoring.

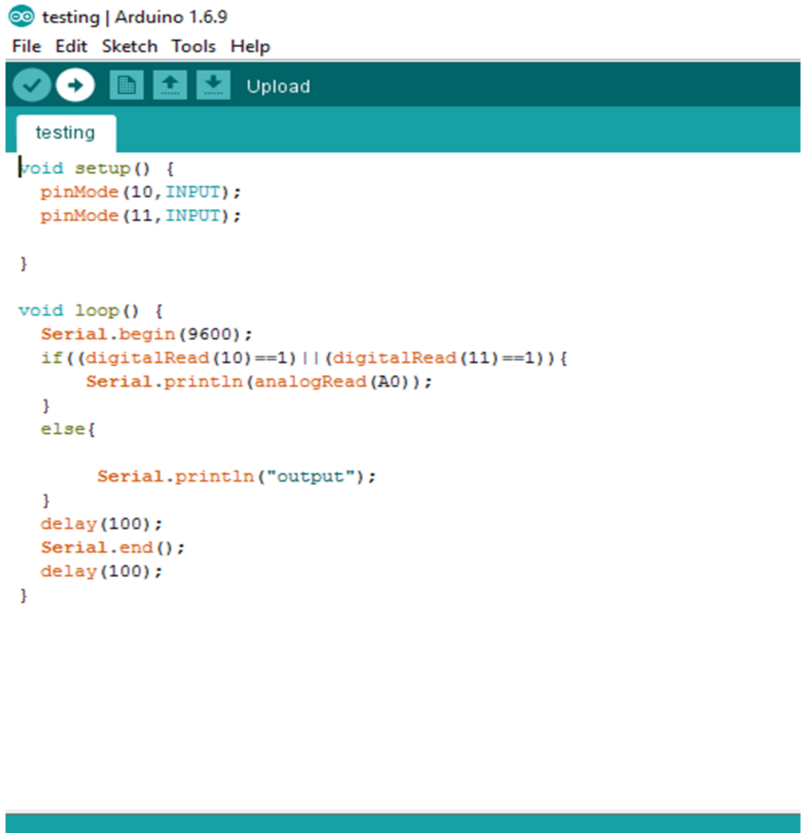

Figure 4-3 ECG Serial monitoring Program

Figure 4-4 ECG Serial monitor Output

4.2.5 Output from the Oscilloscope

We can check our wave form at oscilloscope. It gives us the ECG that we obtained from the analog signal from the electrodes.

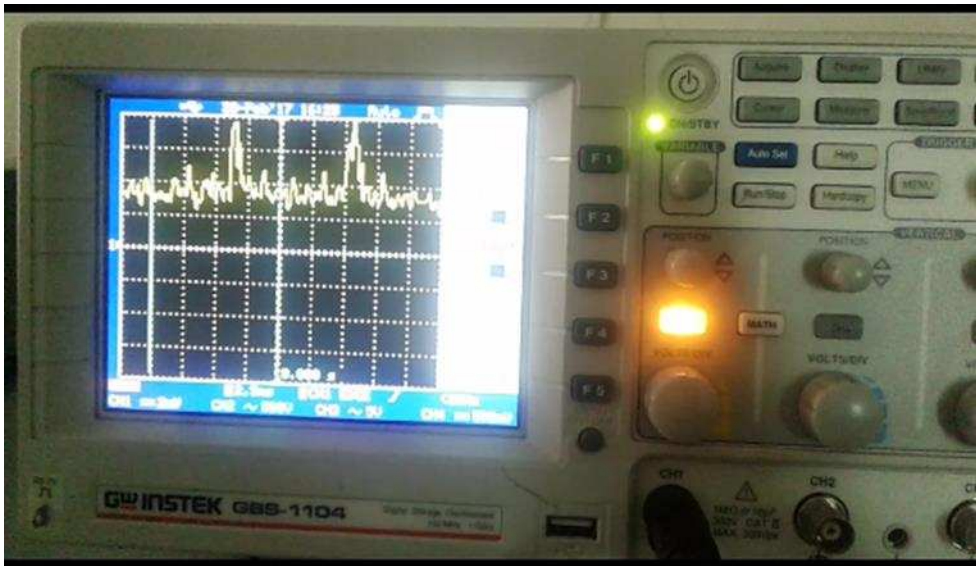

Figure 4-5 Output from the Oscilloscope

4.3 Software Testing

For software testing, normal ECG values and those from the MIT/BIH arrhythmia database were fed into Arduino. During first iteration of the loop, normal ECG values were processed and a normal heart rate of 80 showed up on serial monitor keeping the pin attached to relay of defibrillator low and hence relay maintained its state. In the second iteration and so forth, ECG values from arrhythmia database were processed resulting in heart rate of 27, thus providing pulse to relay to operate and the capacitor discharged itself which was indicated with the help of flash.

Figure 4-6 Software Testing

bpm resulted from ECG values given during testing are shown. When the value is 60, LED was not glowing. But on 27, 53 and 27 values, digital pin 13 goes HIGH and LED is on.

Chapter 5

5 CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

It is evident from this project work that External Defibrillator Vest can be made from cheap components and easy to operate because it is automatically operated. It can measure and observe the ECG (Electrocardiogram) of a patient and on the abnormal ECG, it can delivers shock to the patient to restore its heart beat. The components are highly insulator in order to avoid any charge and leakage and components are packaged into a Vest.

The designed External Defibrillator Vest was tested a number of times and it can operate in the event of disturbed ECG and deliver shock. As this Vest is a prototype and can't be tested on a human body therefore ECG Simulator is employed for testing.

Finally, this External Defibrillator implemented and control part done by Arduino and operate on fibrillation. Hence, this system is scalable and flexible.

5.2 Recommendations

In consonance with the project work and in view of the researched methods and undertakings in the project design, it is recommended to use Arduino Leonardo to improve memory requirements. Due to prototype issues and inadequate facilities of high voltage protection in testing labs, this Vest is restricted to certain voltage levels and energy. Under high voltage protection the voltages can be step up and our prototype shifted to original Vest.

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